Seminar 5:

Classroom problems and worked solutions

25117 - Econometrics

Universitat Pompeu Fabra

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The issue: what changes when we add controls?

- Compare wages and test scores among individuals or schools with similar observed backgrounds.
- Fit a multiple-regression equation and track how additional regressors change predictions and model fit.
- Distinguish an adjusted association from the assumptions needed for a causal comparison.

Classroom problems

Show the formula before substituting numbers. For every controlled comparison, state which regressors are held fixed.

Problem 1: Why do private graduates earn more?

Hourly wages Y (euros per hour) follow the structural model

$$Y = 10 + 4D + 2A + U, \qquad \mathbb{E}(U \mid D, A) = 0,$$

where $D = 1$ denotes a private-university graduate, $D = 0$ a public-university graduate, and A is a parental-education index. Assume finite second moments and

$$\Pr(D = 1) = .5, \qquad \mathbb{E}(A \mid D = 0) = 1, \qquad \mathbb{E}(A \mid D = 1) = 3.$$

1. Find each group's mean wage and the raw private–public wage gap.
2. For the *population* short regression of Y on an intercept and D, calculate $\mathrm{Var}(D)$, $\mathrm{Cov}(D, A)$, and the slope α_1. Compare it with the coefficient on D in the structural model.
3. State the two conditions for omitting A to change this population slope. What if the coefficient on A were zero, or A were uncorrelated with D?

Solution 1: Group means and the omitted-background covariance

1. Iterated expectations give $\mathbb{E}(U \mid D) = 0$, so

$$\mathbb{E}(Y \mid D = d) = 10 + 4d + 2\mathbb{E}(A \mid D = d).$$

$$\mathbb{E}(Y \mid D = 0) = 10 + 2(1) = 12, \qquad \mathbb{E}(Y \mid D = 1) = 10 + 4 + 2(3) = 20.$$

The raw wage gap is $20 - 12 = \boxed{8 \text{ euros per hour}}$.

2. For a binary D, $D^2 = D$. Therefore

$$\mathbb{E}(D) = .5, \qquad \mathrm{Var}(D) = \mathbb{E}(D^2) - [\mathbb{E}(D)]^2 = .5 - .5^2 = \boxed{.25}.$$

$$\mathbb{E}(A) = \sum_{d=0}^{1} \mathbb{E}(A \mid D = d)\Pr(D = d) = .5(1) + .5(3) = 2,$$

$$\mathbb{E}(DA) = \Pr(D = 1)\mathbb{E}(A \mid D = 1) = .5(3) = 1.5,$$

$$\mathrm{Cov}(D, A) = \mathbb{E}(DA) - \mathbb{E}(D)\mathbb{E}(A) = 1.5 - .5(2) = \boxed{.5}.$$

Solution 1: The omitted-variable component (continued)

2. For $Y = \beta_0 + \beta_D D + \beta_A A + U$, $\mathbb{E}(U \mid D, A) = 0$ implies $\mathrm{Cov}(D, U) = 0$. The short-regression population slope is

$$\alpha_1 = \frac{\mathrm{Cov}(D, Y)}{\mathrm{Var}(D)} = \beta_D + \beta_A \frac{\mathrm{Cov}(D, A)}{\mathrm{Var}(D)}$$
$$= 4 + 2\left(\frac{.5}{.25}\right) = \boxed{8}.$$

The coefficient holding A fixed is 4; the raw population slope is 8. The omitted-background component is $8 - 4 = 4$ euros per hour. This comparison concerns population coefficients, rather than every finite-sample estimate.

3. The omitted-variable term is nonzero when both

1. A helps determine Y holding D fixed: $\beta_A \neq 0$;
2. A and D are correlated: $\mathrm{Cov}(D, A) \neq 0$.

If either $\beta_A = 0$ or $\mathrm{Cov}(D, A) = 0$, then

$$\alpha_1 - \beta_D = \beta_A \frac{\mathrm{Cov}(D, A)}{\mathrm{Var}(D)} = 0.$$

Problem 2: Do comparable students have the same tutoring gap?

D indicates tutoring participation and W is a binary background measure recorded before tutoring. The cell summaries are

	No tutoring: $D = 0$		Tutoring: $D = 1$	
Background	Count	Mean score	Count	Mean score
$W = 0$	40	60	10	64
$W = 1$	10	80	40	84

1. Pool the background groups to find each tutoring group's mean score and the raw gap.
2. Find the tutoring gap within each background group. Then standardize both tutoring groups to a common background mix: 50% with $W = 0$, 50% with $W = 1$. Find the two standardized means and their gap.
3. Decompose the raw gap into the within-background gap and a composition term. Do these counts and means establish a causal tutoring effect?

Solution 2: Pooled means and a common background mix

1. Let n_{dw} and $\bar{y}_{dw}$ denote a cell's count and mean. Pool using each tutoring group's own background weights:

$$\bar{y}_d = \sum_{w=0}^{1} \frac{n_{dw}}{n_d} \bar{y}_{dw}, \qquad n_0 = n_1 = 50.$$

$$\bar{y}_0 = \frac{40(60) + 10(80)}{50} = 64, \qquad \bar{y}_1 = \frac{10(64) + 40(84)}{50} = 80.$$

The raw gap is $\bar{y}_1 - \bar{y}_0 = 80 - 64 = \boxed{16 \text{ points}}$.

2. Within background group w, the gap is $\bar{y}_{1w} - \bar{y}_{0w}$:

$$W = 0: \ 64 - 60 = 4, \qquad W = 1: \ 84 - 80 = 4.$$

For common weights $q_0 = q_1 = .5$, use $\bar{y}_d^{\text{std}} = \sum_w q_w \bar{y}_{dw}$:

$$\bar{y}_0^{\text{std}} = .5(60) + .5(80) = 70, \qquad \bar{y}_1^{\text{std}} = .5(64) + .5(84) = 74.$$

The standardized gap is $74 - 70 = \boxed{4 \text{ points}}$.

Solution 2: Within-group differences and composition (continued)

3. The four cell means follow the additive relation

$$\bar{y}_{dw} = 60 + 4d + 20w.$$

The background shares differ between the tutoring groups:

$$\bar{w}_d = \frac{n_{d1}}{n_d}, \qquad \bar{w}_0 = \frac{10}{50} = .2, \quad \bar{w}_1 = \frac{40}{50} = .8.$$

Pooling the cell means gives $\bar{y}_d = 60 + 4d + 20\bar{w}_d$. Therefore

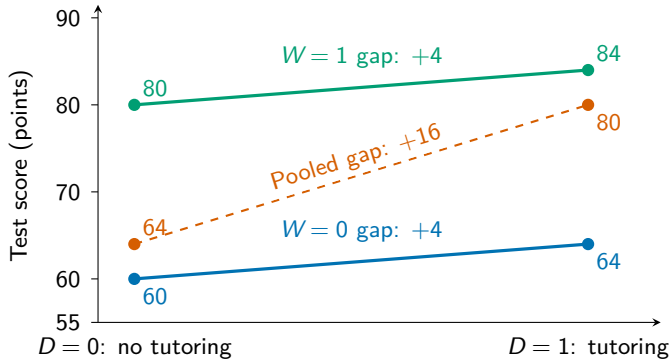
$$\begin{aligned} \bar{y}_1 - \bar{y}_0 &= 4 + 20(\bar{w}_1 - \bar{w}_0) \\ &= \underbrace{4}_{\text{within-background gap}} + \underbrace{20(.8 - .2)}_{\text{composition difference}} = \boxed{16 \text{ points}}. \end{aligned}$$

What the comparison establishes

Reweighting removes the difference in observed background composition from this comparison. The counts and means do not establish random assignment; students can still differ in unobserved determinants of scores within each background group. A causal interpretation needs additional evidence.

Solution 2: See the within-group and pooled gaps (continued)

3. The same tutoring comparison looks different when background groups are pooled:



The pooled line uses each tutoring group's own background mix. Each background-specific line compares students with the same W.

Problem 3: What does holding other regressors fixed mean?

An estimated school-score equation is

$$\widehat{y} = 500 - 1.2M + 4E + .5I,$$

where M is the meal-subsidy share in percentage points, E is average adult schooling in years, and I is annual household income in thousands of euros. Scores are measured in points.

1. Predict scores for school A, $(M, E, I) = (40, 12, 30)$, and school B, $(50, 12, 30)$. Which regressors are held fixed?
2. Predict the score for school C, $(50, 14, 34)$. Decompose the predicted change from A to C into the contributions of M, E, and I. Compare it with the A-to-B change.
3. State the units and ceteris paribus interpretation of each slope. How many slope coefficients and how many total coefficients are estimated?

Solution 3: Partial comparisons and simultaneous changes

1. Substitute into $\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_M M + \widehat{\beta}_E E + \widehat{\beta}_I I$:

$$\widehat{y}_A = 500 - 1.2(40) + 4(12) + .5(30) = \boxed{515},$$
$$\widehat{y}_B = 500 - 1.2(50) + 4(12) + .5(30) = \boxed{503}.$$

Adult schooling and income are held fixed; $\widehat{y}_B - \widehat{y}_A = -12$ points.

2. For school C, $\widehat{y}_C = 500 - 1.2(50) + 4(14) + .5(34) = \boxed{513}$. The change is

$$\begin{aligned}\Delta\widehat{y} &= \widehat{\beta}_M \Delta M + \widehat{\beta}_E \Delta E + \widehat{\beta}_I \Delta I \\ &= -1.2(10) + 4(2) + .5(4) = -12 + 8 + 2 = \boxed{-2 \text{ points}}.\end{aligned}$$

3. Each slope describes a fitted association holding the other two regressors fixed:

Slope	Units and interpretation
-1.2 on M	-1.2 score points per percentage point of meal share
4 on E	$+4$ score points per additional year of adult schooling
.5 on I	$+.5$ score points per additional 1,000 euros of income

The intercept is 500 points. There are $k = 3$ slopes and $k + 1 = 4$ coefficients in total. These fitted associations alone do not establish causality.

Problem 4: Which conditions must a fitted plane satisfy?

Six observations on two regressors and an outcome are

Observation i	1	2	3	4	5	6
x_i	0	0	1	1	2	2
w_i	0	1	0	1	0	1
y_i	11	12	10	17	15	16

Fit a plane $\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i + \widehat{\beta}_2 w_i$.

1. Define the squared-error objective $Q(b_0, b_1, b_2)$. Use the chain rule to derive its three first-order conditions, showing the derivative of a residual with respect to each coefficient.
2. For the candidate $(b_0, b_1, b_2) = (10, 2, 3)$, calculate fitted values and residuals. Verify $\sum_i \widehat{u}_i = 0$, $\sum_i x_i \widehat{u}_i = 0$, and $\sum_i w_i \widehat{u}_i = 0$.
3. Derive the intercept relation from the first condition. Verify it using the sample means and the candidate slopes.

Solution 4: Chain-rule derivatives and three normal equations

1. For a candidate coefficient vector $\mathbf{b} = (b_0, b_1, b_2)'$, define

$$r_i(\mathbf{b}) = y_i - b_0 - b_1 x_i - b_2 w_i, \qquad Q(\mathbf{b}) = \sum_{i=1}^{n} r_i(\mathbf{b})^2.$$

The residual derivatives are

$$\frac{\partial r_i}{\partial b_0} = -1, \qquad \frac{\partial r_i}{\partial b_1} = -x_i, \qquad \frac{\partial r_i}{\partial b_2} = -w_i.$$

Apply the chain rule, $\partial Q/\partial b_j = \sum_i 2 r_i(\mathbf{b})\, \partial r_i/\partial b_j$, and set the derivatives to zero at the minimum:

$$\frac{\partial Q}{\partial b_0} = -2 \sum_i (y_i - b_0 - b_1 x_i - b_2 w_i) = 0,$$

$$\frac{\partial Q}{\partial b_1} = -2 \sum_i x_i (y_i - b_0 - b_1 x_i - b_2 w_i) = 0,$$

$$\frac{\partial Q}{\partial b_2} = -2 \sum_i w_i (y_i - b_0 - b_1 x_i - b_2 w_i) = 0.$$

At OLS, evaluate these equations at $\mathbf{b} = \widehat{\boldsymbol{\beta}}$.

Solution 4: Check residuals and recover the intercept (continued)

2. Use $\widehat{y}_i = 10 + 2x_i + 3w_i$ and $\widehat{u}_i = y_i - \widehat{y}_i$:

i	x_i	w_i	y_i	$\widehat{y}_i$	$\widehat{u}_i$
1	0	0	11	10	1
2	0	1	12	13	−1
3	1	0	10	12	−2
4	1	1	17	15	2
5	2	0	15	14	1
6	2	1	16	17	−1

$$\sum_i \widehat{u}_i = 1 - 1 - 2 + 2 + 1 - 1 = 0,$$

$$\sum_i x_i \widehat{u}_i = 0 + 0 - 2 + 2 + 2 - 2 = 0,$$

$$\sum_i w_i \widehat{u}_i = 0 - 1 + 0 + 2 + 0 - 1 = 0.$$

3. The means are $\overline{x} = 6/6 = 1$, $\overline{w} = 3/6 = .5$, and $\overline{y} = 81/6 = 13.5$. Divide the intercept condition by n:

$$\sum_i \widehat{u}_i = 0 \quad \Longrightarrow \quad \overline{y} - \widehat{\beta}_0 - \widehat{\beta}_1 \overline{x} - \widehat{\beta}_2 \overline{w} = 0.$$

$$\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x} - \widehat{\beta}_2 \overline{w} = 13.5 - 2(1) - 3(.5) = \boxed{10}.$$

The normal equations are fitting identities; they do not establish causality.

Problem 5: How do the scalar conditions become matrices?

Use the six observations in Problem 4 and the coefficient order (b_0, b_1, b_2) for the intercept, x, and w.

1. Form the design matrix $\mathbf{X}$, outcome vector $\mathbf{Y}$, and coefficient vector $\mathbf{b}$. Include the intercept column and state all dimensions. What are the dimensions of $\mathbf{Xb}$ and $\mathbf{Y} - \mathbf{Xb}$?

2. Compute $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}'\mathbf{Y}$, showing which sample sums form their entries. State their dimensions.

3. Write the three scalar normal equations represented by $(\mathbf{X}'\mathbf{X})\mathbf{b} = \mathbf{X}'\mathbf{Y}$. Match each row to a first-order condition from Problem 4. Write the gradient of Q in matrix form and check its dimensions.

Solution 5: Stack observations and keep the intercept column

1. Each row of the design matrix is $(1, x_i, w_i)$:

$$\mathbf{X}_{6\times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 2 & 1 \end{bmatrix}, \qquad \mathbf{Y}_{6\times 1} = \begin{bmatrix} 11 \\ 12 \\ 10 \\ 17 \\ 15 \\ 16 \end{bmatrix}, \qquad \mathbf{b}_{3\times 1} = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix}.$$

The first column of ones makes every row's fitted value include the intercept:

$$(\mathbf{Xb})_i = 1b_0 + x_i b_1 + w_i b_2.$$

Check the matrix multiplication and subtraction:

$$\underbrace{\mathbf{X}}_{6\times 3}\underbrace{\mathbf{b}}_{3\times 1} = \underbrace{\mathbf{Xb}}_{6\times 1}, \qquad \underbrace{\mathbf{Y} - \mathbf{Xb}}_{6\times 1} = \begin{bmatrix} r_1(\mathbf{b}) \\ \vdots \\ r_6(\mathbf{b}) \end{bmatrix}.$$

Solution 5: Cross-products of the data (continued)

2. Compute the cross-product matrices from the sample sums:

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} n & \sum_i x_i & \sum_i w_i \\ \sum_i x_i & \sum_i x_i^2 & \sum_i x_i w_i \\ \sum_i w_i & \sum_i x_i w_i & \sum_i w_i^2 \end{bmatrix} = \underbrace{\begin{bmatrix} 6 & 6 & 3 \\ 6 & 10 & 3 \\ 3 & 3 & 3 \end{bmatrix}}_{3\times 3},$$

$$\mathbf{X}'\mathbf{Y} = \begin{bmatrix} \sum_i y_i \\ \sum_i x_i y_i \\ \sum_i w_i y_i \end{bmatrix} = \underbrace{\begin{bmatrix} 81 \\ 89 \\ 45 \end{bmatrix}}_{3\times 1}.$$

Solution 5: Normal equations and gradient (continued)

3. The matrix equation $(\mathbf{X}'\mathbf{X})\mathbf{b} = \mathbf{X}'\mathbf{Y}$ contains

$$6b_0 + 6b_1 + 3b_2 = 81 \qquad (\text{intercept: } \sum_i r_i = 0),$$

$$6b_0 + 10b_1 + 3b_2 = 89 \qquad (x\text{: } \sum_i x_i r_i = 0),$$

$$3b_0 + 3b_1 + 3b_2 = 45 \qquad (w\text{: } \sum_i w_i r_i = 0).$$

For $Q(\mathbf{b}) = (\mathbf{Y} - \mathbf{X}\mathbf{b})'(\mathbf{Y} - \mathbf{X}\mathbf{b})$,

$$\nabla_{\mathbf{b}} Q = -2 \underbrace{\mathbf{X}'}_{3\times 6} \underbrace{(\mathbf{Y} - \mathbf{X}\mathbf{b})}_{6\times 1} \quad \text{is } 3 \times 1.$$

Setting this gradient to zero recovers all three scalar conditions.

Problem 6: How do we solve the multiple-regression equations?

Use the design matrix and scores from Problems 4–5, with columns ordered as the intercept, X, and W:

$$G = \mathbf{X}'\mathbf{X} = \begin{pmatrix} 6 & 6 & 3 \\ 6 & 10 & 3 \\ 3 & 3 & 3 \end{pmatrix}, \qquad h = \mathbf{X}'\mathbf{Y} = \begin{pmatrix} 81 \\ 89 \\ 45 \end{pmatrix}.$$

A proposed inverse is

$$A = \begin{pmatrix} 7/12 & -1/4 & -1/3 \\ -1/4 & 1/4 & 0 \\ -1/3 & 0 & 2/3 \end{pmatrix}.$$

1. Verify that $GA = I_3$. What does invertibility tell us about the design matrix?
2. Solve the normal equations $G\widehat{\boldsymbol{\beta}} = h$ by multiplying Ah, showing each coefficient calculation.
3. Predict scores at $x = 1.5$ for $w = 0$ and $w = 1$. Interpret the coefficient on W while holding X fixed.

Solution 6: Checking the inverse and full column rank

1. Multiply the proposed inverse by G on the left:

$$GA = \begin{pmatrix} \frac{7}{2} - \frac{3}{2} - 1 & -\frac{3}{2} + \frac{3}{2} & -2 + 2 \\ \frac{7}{2} - \frac{5}{2} - 1 & -\frac{3}{2} + \frac{5}{2} & -2 + 2 \\ \frac{7}{4} - \frac{3}{4} - 1 & -\frac{3}{4} + \frac{3}{4} & -1 + 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I_3.$$

Therefore $A = G^{-1}$. Since $G = \mathbf{X}'\mathbf{X}$ is invertible, the three columns of $\mathbf{X}$ are linearly independent: the design has full column rank.

The normal equations consequently have the unique solution

$$G\widehat{\boldsymbol{\beta}} = h \quad \Longrightarrow \quad \widehat{\boldsymbol{\beta}} = G^{-1}h = Ah.$$

Solution 6: The coefficients and controlled predictions

2. For $j = 0, 1, 2$, $\widehat{\beta}_j = \sum_{\ell=1}^{3} A_{j+1,\ell} h_\ell$. Thus

$$\widehat{\beta}_0 = \frac{7}{12}(81) - \frac{1}{4}(89) - \frac{1}{3}(45) = 47.25 - 22.25 - 15 = \boxed{10},$$

$$\widehat{\beta}_1 = -\frac{1}{4}(81) + \frac{1}{4}(89) + 0(45) = \boxed{2},$$

$$\widehat{\beta}_2 = -\frac{1}{3}(81) + 0(89) + \frac{2}{3}(45) = \boxed{3}.$$

3. Use $\widehat{y}(x, w) = \widehat{\beta}_0 + \widehat{\beta}_1 x + \widehat{\beta}_2 w$:

$$\widehat{y}(1.5, 0) = 10 + 2(1.5) + 3(0) = 13,$$

$$\widehat{y}(1.5, 1) = 10 + 2(1.5) + 3(1) = 16.$$

Holding X fixed, changing W from 0 to 1 is associated with a fitted score increase of 3 points. This fitted comparison does not establish causality by itself.

Problem 7: How well does the fitted plane describe scores?

Return to the six observations in Problems 4–6. The model has an intercept and two regressors, and

$$n = 6, \qquad SSR = 12, \qquad TSS_Y = 41.5.$$

1. Count the estimated coefficients and residual degrees of freedom. Compute *SER* and the in-sample *RMSE*.
2. Find the explained sum of squares, *ESS*, and calculate R^2.
3. Calculate adjusted R^2 using both the ratio of corrected sums of squares and the expression involving $1 - R^2$.
4. Explain why adjusted R^2 is no larger than R^2 here. State the units of *SER* and R^2 and explain what each measures.

Solution 7: Residual size and the unadjusted fit

1. Here $k = 2$ counts regressors other than the intercept:

$$\text{coefficient count} = k + 1 = 3, \qquad \text{residual degrees of freedom} = n - k - 1 = 6 - 2 - 1 = 3.$$

The two residual-size summaries are

$$SER = \sqrt{\frac{SSR}{n-k-1}} = \sqrt{\frac{12}{3}} = \sqrt{4} = \boxed{2 \text{ points}},$$

$$RMSE = \sqrt{\frac{SSR}{n}} = \sqrt{\frac{12}{6}} = \sqrt{2} \approx 1.4142 \text{ points}.$$

2. With an intercept, the sums-of-squares decomposition gives

$$ESS = TSS_Y - SSR = 41.5 - 12 = 29.5,$$

$$R^2 = 1 - \frac{SSR}{TSS_Y} = 1 - \frac{12}{41.5} = \frac{29.5}{41.5} = \boxed{\frac{59}{83} \approx .7108}.$$

Solution 7: Adjusted fit and units (continued)

3. First compare the corrected sums of squares:

$$\overline{R}^2 = 1 - \frac{SSR/(n-k-1)}{TSS_Y/(n-1)} = 1 - \frac{12/3}{41.5/5} = 1 - \frac{4}{8.3} = \boxed{\frac{43}{83} \approx .5181}.$$

The equivalent formula gives the same value:

$$\overline{R}^2 = 1 - \frac{n-1}{n-k-1}(1-R^2) = 1 - \frac{5}{3}\left(1 - \frac{59}{83}\right) = 1 - \frac{40}{83} = \frac{43}{83}.$$

4. The correction multiplies the unexplained share $1 - R^2$ by $5/3 > 1$, so $\overline{R}^2 < R^2$ here.

$SER = 2$ is measured in score points and summarizes residual size with a degrees-of-freedom correction. R^2 is unitless: the fitted plane accounts for about 71.1% of the observed variation around the sample mean.

These fit measures alone do not provide a formal prediction interval or establish a causal effect.

Problem 8: Does adding a control improve the adjusted fit?

Three OLS models use the same 40 observations and outcome, with $TSS_Y = 200$. Each model includes an intercept and has full column rank.

Model	Regressor count k	Comparison with A	SSR
A	1	Initial specification	100
B	2	Adds one control	98
C	2	Adds a different control	90

B and C each nest A; B and C do not nest each other.

1. Calculate R^2 and adjusted R^2 for all three models.
2. Derive how small the new SSR must be for a one-control expansion of A to improve adjusted R^2.
3. Explain why adding a control weakly raises raw R^2 but can lower adjusted R^2. Can these statistics alone determine the controls needed for a causal analysis?

Solution 8: Comparing raw and adjusted fit

1. For each model,

$$R^2 = 1 - \frac{SSR}{TSS_Y}, \qquad \overline{R}^2 = 1 - \frac{SSR/(n-k-1)}{TSS_Y/(n-1)}.$$

With $TSS_Y = 200$ and $n = 40$:

Model	k	SSR	$n-k-1$	R^2	$\overline{R}^2$
A	1	100	38	.5000	.4868
B	2	98	37	.5100	.4835
C	2	90	37	.5500	.5257

For example, the adjusted calculations are

$$\overline{R}^2_A = 1 - \frac{100/38}{200/39} = \frac{37}{76} \approx .4868,$$

$$\overline{R}^2_B = 1 - \frac{98/37}{200/39} = \frac{1789}{3700} \approx .4835,$$

$$\overline{R}^2_C = 1 - \frac{90/37}{200/39} = \frac{389}{740} \approx .5257.$$

Solution 8: The improvement threshold and control selection

2. The denominator $TSS_Y/(n-1)$ is common to the models. Therefore,

$$\overline{R}^2_{\text{new}} > \overline{R}^2_A \iff \frac{SSR_{\text{new}}}{40-2-1} < \frac{100}{40-1-1}$$
$$\iff SSR_{\text{new}} < 100\frac{37}{38} \approx \boxed{97.3684}.$$

B's $SSR = 98$ fails this threshold; C's $SSR = 90$ satisfies it.

3. In an expanded model, setting the new coefficient to zero reproduces the old fitted values. On the same observations, minimized SSR can only fall or stay unchanged, so raw R^2 can only rise or stay unchanged.

Adjusted R^2 also accounts for the lost degree of freedom. A small reduction in SSR may not compensate for this penalty, as model B shows.

Causal controls must be chosen using the identification argument and knowledge of the variables. Fit statistics alone cannot determine the appropriate causal specification.

Problem 9: How can adjusted R-squared be negative?

An OLS regression uses $n = 10$ observations, an intercept, and $k = 3$ regressors. The design has full column rank, and

$$TSS_Y = 90, \qquad SSR = 80.$$

Scores are measured in points.

1. Calculate raw R^2 and adjusted R^2.

2. Calculate $RMSE$, SER, and the sample variance s_Y^2. Which comparison explains the sign of adjusted R^2?

3. An intercept-only regression on the same observations has $SSR = 90$. Calculate its adjusted R^2 and compare its corrected residual average with that of the larger model. Why can a model with lower SSR have worse adjusted fit?

Solution 9: Raw fit, adjusted fit, and residual size

1. With $n - k - 1 = 10 - 3 - 1 = 6$,

$$R^2 = 1 - \frac{SSR}{TSS_Y} = 1 - \frac{80}{90} = \boxed{\frac{1}{9} \approx .1111},$$

$$\overline{R}^2 = 1 - \frac{SSR/(n-k-1)}{TSS_Y/(n-1)} = 1 - \frac{80/6}{90/9} = 1 - \frac{4}{3} = \boxed{-\frac{1}{3}}.$$

2. The residual-size summaries and outcome variance are

$$RMSE = \sqrt{\frac{SSR}{n}} = \sqrt{\frac{80}{10}} = \sqrt{8} \approx 2.8284 \text{ points},$$

$$SER = \sqrt{\frac{SSR}{n-k-1}} = \sqrt{\frac{80}{6}} \approx 3.6515 \text{ points}, \qquad s_Y^2 = \frac{TSS_Y}{n-1} = \frac{90}{9} = 10 \text{ points}^2.$$

Because $SER^2 = 80/6 > 10 = s_Y^2$, their ratio exceeds one and adjusted R^2 is negative. Raw explained variation remains nonnegative.

Solution 9: The intercept-only benchmark (continued)

3. The intercept-only model estimates one coefficient, the sample mean. With $k = 0$, its residual degrees of freedom are $n - 1 = 9$:

$$\overline{R}^2_{\text{intercept}} = 1 - \frac{SSR/(n-1)}{TSS_Y/(n-1)} = 1 - \frac{90/9}{90/9} = \boxed{0}.$$

Compare the corrected residual averages:

$$\frac{SSR_{\text{intercept}}}{9} = \frac{90}{9} = 10, \qquad \frac{SSR_{\text{larger}}}{6} = \frac{80}{6} = 13\frac{1}{3}.$$

The larger model reduces SSR by 10, but uses three additional coefficients. Its reduction in squared residuals is too small to offset the loss of residual degrees of freedom.

A negative adjusted R^2 is possible because it is a penalized fit statistic. This comparison alone does not determine the controls needed for a causal analysis.

Problem 10: What do the coefficient variances tell us?

A regression using a large i.i.d. sample of $n = 200$ observations estimates

$$Y = \beta_0 + \beta_1 X + \beta_2 W + U, \qquad \widehat{\boldsymbol{\beta}} = \begin{pmatrix} 2 \\ 1.5 \\ -.8 \end{pmatrix}.$$

An appropriate estimated coefficient covariance matrix, ordered as the intercept, X, and W, is supplied:

$$\widehat{V} = \begin{pmatrix} .25 & .02 & -.01 \\ .02 & .36 & .04 \\ -.01 & .04 & .16 \end{pmatrix}.$$

Use the large-sample normal approximation, $z_{.975} = 1.96$, $\Phi(2.5) = .9938$, and $\Phi(2) = .97725$.

1. Read off the estimated variances and covariances, and calculate all three coefficient standard errors.
2. Test each slope separately against zero using a two-sided 5% test. Compute both p-values.
3. Construct the two 95% slope confidence intervals. Relate zero's membership to the tests, and state what the results establish.

Solution 10: Reading variances and forming separate tests

1. For coefficient j, $\widehat{\text{Var}}(\widehat{\beta}_j) = \widehat{V}_{j+1,j+1}$ and $SE(\widehat{\beta}_j) = \sqrt{\widehat{V}_{j+1,j+1}}$. Thus

Coefficient	$\widehat{\beta}_0$	$\widehat{\beta}_1$	$\widehat{\beta}_2$
Estimated variance	.25	.36	.16
Standard error	$\sqrt{.25} = .50$	$\sqrt{.36} = .60$	$\sqrt{.16} = .40$

The estimated covariances are

$$\widehat{\text{Cov}}(\widehat{\beta}_0, \widehat{\beta}_1) = .02, \quad \widehat{\text{Cov}}(\widehat{\beta}_0, \widehat{\beta}_2) = -.01, \quad \widehat{\text{Cov}}(\widehat{\beta}_1, \widehat{\beta}_2) = .04.$$

2. For each $j = 1, 2$, test $H_0 : \beta_j = 0$ against $H_1 : \beta_j \neq 0$, using

$$t_j = \frac{\widehat{\beta}_j - 0}{SE(\widehat{\beta}_j)}, \qquad p_j \approx 2[1 - \Phi(|t_j|)].$$

$$t_1 = \frac{1.5}{.60} = 2.5, \quad p_1 \approx 2(1 - .9938) = .0124,$$
$$t_2 = \frac{-.8}{.40} = -2, \quad p_2 \approx 2(1 - .97725) = .0455.$$

Both separate nulls are rejected at 5%: $|t_j| > 1.96$ and $p_j < .05$.

Solution 10: Slope confidence intervals and interpretation

3. For each slope, the large-sample 95% interval is

$$\widehat{\beta}_j \pm 1.96\, SE(\widehat{\beta}_j).$$

For the coefficient on X,

$$1.5 \pm 1.96\sqrt{.36} = 1.5 \pm 1.96(.60) = 1.5 \pm 1.176 = \boxed{[.324,\ 2.676]}.$$

For the coefficient on W,

$$-.8 \pm 1.96\sqrt{.16} = -.8 \pm 1.96(.40) = -.8 \pm .784 = \boxed{[-1.584,\ -.016]}.$$

Neither interval contains zero, agreeing with rejection in each corresponding two-sided 5% test.

The estimated association with X is positive holding W fixed; the estimated association with W is negative holding X fixed. These are separate coefficient tests and intervals, not a joint test. A causal interpretation requires an identification argument beyond these results.

Problem 11: Can two measures of study time have separate coefficients?

Regress Y on an intercept, study hours X_1, and study minutes X_2:

x_{1i}	1	2	3	4	5
x_{2i}	60	120	180	240	300
y_i	15	20	25	30	35

1. Show the exact relation among the regressor columns. Why is the design matrix's $\mathbf{X}'\mathbf{X}$ not invertible?
2. Which combination of the two slope coefficients is identified? Check that $(b_1, b_2) = (5, 0)$, $(2, .05)$, and $(-1, .1)$ give the same fitted values when $b_0 = 10$. What happens if we drop study minutes?
3. Replace the minutes data by $(60, 121, 179, 241, 300)$. Is there still an exact affine relation with hours? Does the new design have full column rank? Distinguish this case from perfect multicollinearity and count the residual degrees of freedom.

Solution 11: One dimension of study time, two columns

1. The columns satisfy $x_{2i} - 60x_{1i} = 0$ for every observation. For the design matrix $\mathbf{X} = [\mathbf{1}, \mathbf{x}_1, \mathbf{x}_2]$,

$$v = \begin{pmatrix} 0 \\ -60 \\ 1 \end{pmatrix} \neq 0, \qquad \mathbf{X}v = 0 \quad \Longrightarrow \quad \mathbf{X}'\mathbf{X}v = 0.$$

Thus $\mathbf{X}'\mathbf{X}$ has a nonzero null vector and is not invertible.

2. Substitute the column relation into any candidate fitted line:

$$b_0 + b_1 x_1 + b_2 x_2 = b_0 + (b_1 + 60b_2)x_1.$$

The data identify $b_0 = 10$ and $\boxed{b_1 + 60b_2 = 5}$. They do not identify the two slopes separately:

$$5 + 60(0) = 5, \qquad 2 + 60(.05) = 5, \qquad -1 + 60(.1) = 5.$$

Each coefficient pair produces $\widehat{y} = 10 + 5x_1$ and the same zero residuals. Dropping the minutes regressor gives the unique simple-regression fit $\boxed{\widehat{y} = 10 + 5x_1}$.

Solution 11: Near collinearity permits separate coefficients (continued)

3. Suppose the new minutes column satisfied $x_2 = a + bx_1$. Its first and fifth observations would imply

$$60 = a + b, \qquad 300 = a + 5b \quad \Longrightarrow \quad b = 60,\ a = 0.$$

But at the second observation this relation predicts $x_2 = 120$, whereas the new value is 121. Hence no exact affine relation holds.

Hours are nonconstant, and minutes cannot be written as a constant plus hours. The three columns therefore have full column rank. With three coefficients,

$$\text{residual degrees of freedom} = n - 3 = 5 - 3 = \boxed{2}.$$

The columns remain extremely similar, so this is **near collinearity**, rather than perfect multicollinearity.

High correlation among regressors does not itself create bias. In data with noise, however, it can make the separate coefficient estimates imprecise and sensitive to small changes. In these particular data, $\widehat{y} = 10 + 5x_1 + 0x_2$ still fits exactly.

Problem 12: Which category should be the reference group?

A sample contains four graduates from each of three university categories. Every graduate belongs to exactly one category. Their mean hourly wages are

Category	Public	Private	Other
Mean wage (euros/hour)	18	24	20

Let P, R, and O be the corresponding category dummies.

1. Why is a regression on an intercept and all three dummies not identified?
2. Omit the public dummy. Find the intercept and the coefficients on the private and other dummies.
3. Use private universities as the reference category instead. Find all coefficients and check predictions for the three categories.
4. Omit the intercept and retain all three dummies. Find the coefficients. Explain why these valid specifications have the same fitted values. Do the wage gaps automatically measure causal effects?

Solution 12: A column relation and a public reference group

1. For every graduate, exactly one dummy is one:

$$P_i + R_i + O_i = 1.$$

The intercept column equals the sum of all three dummy columns, creating perfect multicollinearity. Four separate coefficients cannot identify only three group means.

2. For group-specific fitted constants m_g, OLS minimizes

$$\sum_g \sum_{i \in g} (y_i - m_g)^2, \qquad \frac{\partial Q}{\partial m_g} = -2 \sum_{i \in g} (y_i - m_g) = 0 \quad \Longrightarrow \quad m_g = \bar{y}_g.$$

With public omitted, $\widehat{y} = b_0 + b_R R + b_O O$. Therefore

$$\boxed{b_0 = \bar{y}_P = 18}, \qquad \boxed{b_R = \bar{y}_R - \bar{y}_P = 24 - 18 = 6},$$

$$\boxed{b_O = \bar{y}_O - \bar{y}_P = 20 - 18 = 2}.$$

The intercept is the public-group mean; each dummy coefficient is a gap from that reference group, in euros per hour.

Solution 12: Changing the reference preserves fitted wages (continued)

3. With private omitted, $\widehat{y} = c_0 + c_P P + c_O O$:

$$\boxed{c_0 = 24}, \qquad \boxed{c_P = 18 - 24 = -6}, \qquad \boxed{c_O = 20 - 24 = -4}.$$

Category	Public	Private	Other
Prediction	$24 - 6 = 18$	24	$24 - 4 = 20$

4. Without an intercept, write $\widehat{y} = a_P P + a_R R + a_O O$. Each coefficient directly equals its group's fitted mean:

$$\boxed{a_P = 18}, \qquad \boxed{a_R = 24}, \qquad \boxed{a_O = 20}.$$

All three valid parameterizations reproduce the same group means, so they have the same fitted values and residuals. Changing the reference changes the labels and interpretation of coefficients, rather than the fit.

The wage gaps are descriptive. They do not automatically identify causal effects of attending a particular university category.

Problem 13: Can a useful control have a noncausal coefficient?

Consider the structural model

$$Y = 10 + 4D + 2W + U, \qquad \mathbb{E}(U \mid D, W) = 3W,$$

where D is binary participation and W is a binary background characteristic. All four (D, W) cells have positive probability, and variables have finite second moments. The model stipulates that changing D while holding W and U fixed changes Y by 4.

1. Tabulate $\mathbb{E}(U \mid D, W)$ for the four cells. Does conditional mean independence, $\mathbb{E}(U \mid D, W) = \mathbb{E}(U \mid W)$, hold? Does full zero conditional mean hold?
2. Derive $\mathbb{E}(Y \mid D, W)$ and the within-W participation gap. Explain why the binary control absorbs the background-dependent error mean exactly.
3. Which population regression coefficient targets the stipulated causal participation effect? Why need the control coefficient not equal its structural coefficient of 2?

Solution 13: Conditional mean independence is sufficient here

1. The conditional error means do not change with D at a given W:

$\mathbb{E}(U \mid D, W)$	$D=0$	$D=1$
$W=0$	0	0
$W=1$	3	3

Iterated expectations gives $\mathbb{E}(U \mid W) = \mathbb{E}[3W \mid W] = 3W$. Hence

$$\boxed{\mathbb{E}(U \mid D, W) = \mathbb{E}(U \mid W)}.$$

Conditional mean independence holds, but full zero conditional mean fails at $W = 1$.

2. Taking conditional expectations of the structural equation,

$$\begin{aligned}\mathbb{E}(Y \mid D, W) &= 10 + 4D + 2W + \mathbb{E}(U \mid D, W) \\ &= 10 + 4D + 2W + 3W = \boxed{10 + 4D + 5W}.\end{aligned}$$

For either $w = 0$ or $w = 1$, the participation gap is

$$\mathbb{E}(Y \mid D = 1, W = w) - \mathbb{E}(Y \mid D = 0, W = w) = \boxed{4}.$$

Solution 13: The participation and control coefficients differ (continued)

2. An intercept and binary W can represent any function of W:

$$g(W) = g(0) + [g(1) - g(0)]W.$$

Here the omitted-factor mean $g(W) = 3W$ is absorbed exactly by the control coefficient. Define $V = U - 3W$; then

$$Y = 10 + 4D + 5W + V, \qquad \mathbb{E}(V \mid D, W) = 0.$$

3. The population regression coefficient on D is $\boxed{4}$, which equals the causal participation effect stipulated by this model. Conditioning on W removes the participation-related difference in the mean of U.

The population coefficient on W is instead

$$\boxed{5} = \underbrace{2}_{\text{structural coefficient}} + \underbrace{3}_{\text{background-linked error mean}}.$$

That control coefficient need not measure the causal effect of changing W. A useful control does not make every regression coefficient causal.

The exact absorption above uses binary W. With other controls, the functional form must represent the relevant background-dependent mean adequately.

Problem 14: When does a control fail to remove selection?

Keep the structural model and the causal participation effect of 4, but change the error mean:

$$Y = 10 + 4D + 2W + U, \qquad \mathbb{E}(U \mid D, W) = 3W + 2D.$$

D and W are binary, all four cells have positive probability, and observations are i.i.d. with bounded variables and a full-rank regression design.

1. Tabulate $\mathbb{E}(U \mid D, W)$. At each value of W, does its mean depend on participation? Does conditional mean independence hold?

2. Derive $\mathbb{E}(Y \mid D, W)$. Compare the population multiple-regression coefficient on D with its structural causal effect.

3. Would more observations, finite fourth moments, and full rank repair the discrepancy? Identify the failing condition and explain what further design evidence or pre-treatment controls would be needed for a credible causal interpretation.

Solution 14: Selection remains within each background group

1. The new conditional error means are

$\mathbb{E}(U \mid D, W)$	$D = 0$	$D = 1$	Difference $(D = 1) - (D = 0)$
$W = 0$	0	2	2
$W = 1$	3	5	2

The error mean changes with D even after fixing W. Conditional mean independence therefore fails.

2. Taking conditional expectations,

$$\begin{aligned}\mathbb{E}(Y \mid D, W) &= 10 + 4D + 2W + \mathbb{E}(U \mid D, W) \\ &= 10 + 4D + 2W + 3W + 2D = \boxed{10 + 6D + 5W}.\end{aligned}$$

The population observational coefficient on D is $\boxed{6}$, while the stipulated structural causal effect is $\boxed{4}$. The remaining within-W selection adds 2 to the participation gap.

Solution 14: Selection and sample size (continued)

3. I.i.d. sampling, finite fourth moments, and full rank do not imply

$$\mathbb{E}(U \mid D, W) = \mathbb{E}(U \mid W).$$

This conditional mean independence condition fails in the stated model. More observations estimate the observational coefficient of 6 more closely; they do not change it into the structural effect of 4.

For a causal interpretation, we would need credible evidence that participation no longer selects on omitted determinants of Y within the relevant background groups. A defensible assignment mechanism or sufficiently informative pre-treatment controls could support that argument.

Additional controls help only when they address the source of selection and represent the relevant conditional means adequately. The number of controls is not evidence that the selection problem has been removed.